

Practical Work #5

Parameters of quadripoles

Objectives:

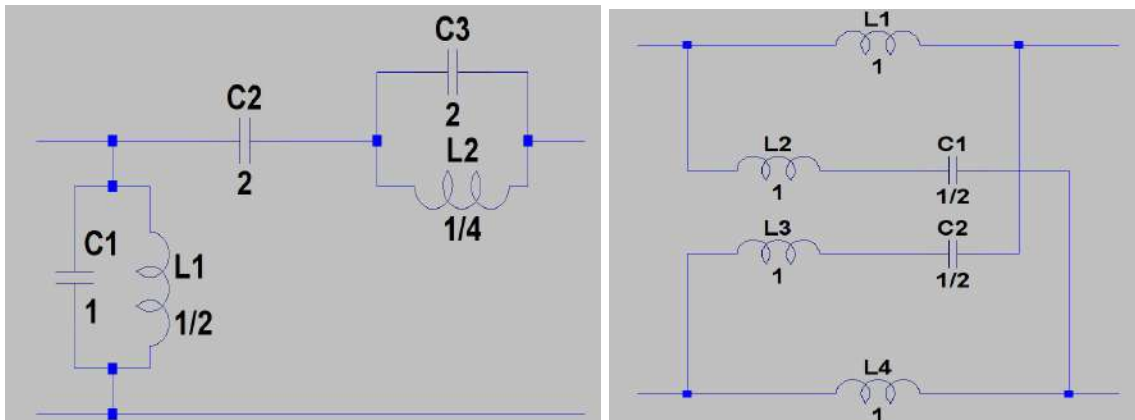
- Analyze electrical networks obtaining their characteristic parameters
- Familiarize with the relationships between the different quadripole parameters
- Recognize by simple inspection the most suitable parameters for analyzing a network.
- Be able to transform balanced quadripoles to unbalanced ones and vice versa.
- Intuitively synthesize a quadripole
- Obtain different networks from the same mathematical model
- Be able to couple quadripoles in their different configurations

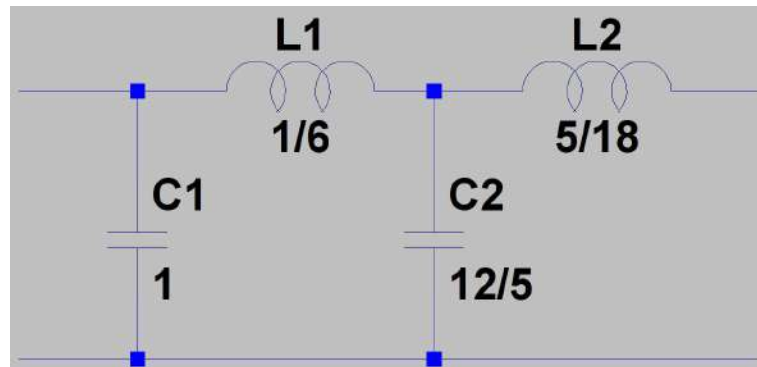
Approval condition: the practical work must be submitted up to 2 weeks after it is carried out in class.

Responsible: each practical work must have a person responsible. Submissions will not be accepted without the corresponding cover.

Exercise #1

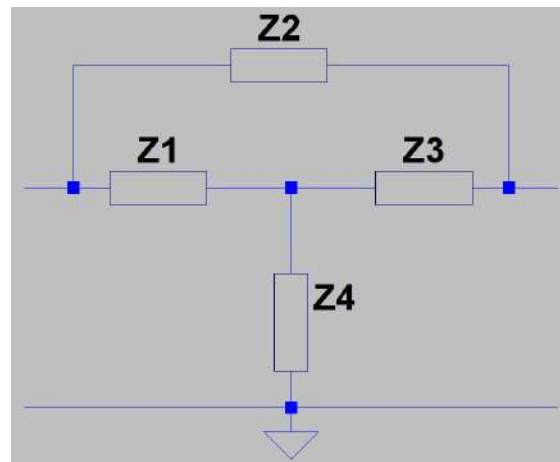
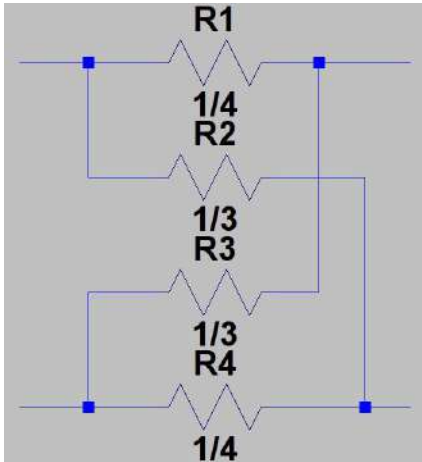
Obtain the Z, Y, and T matrices of the following quadripoles





Exercise #2

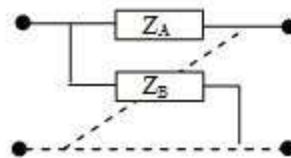
Transform the following lattice quadripole into an equivalent T-bridged one



Exercise #3

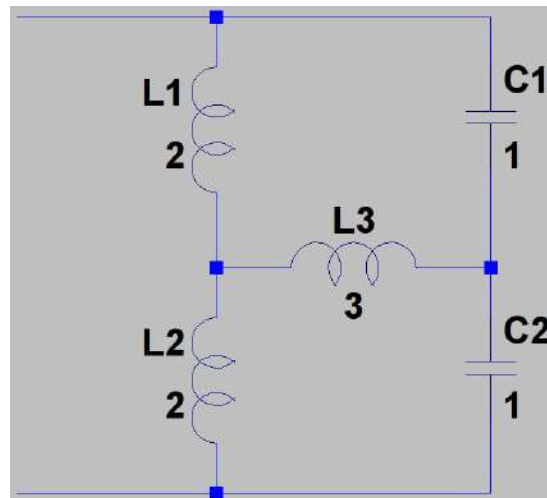
Given the following T-matrix, obtain the components of the lattice quadripole that satisfy it.
 Note: Consider the passive and symmetric quadripole.

$$[T] = \begin{bmatrix} s^2 + 1 & B \\ s & D \end{bmatrix}$$



Exercise #4

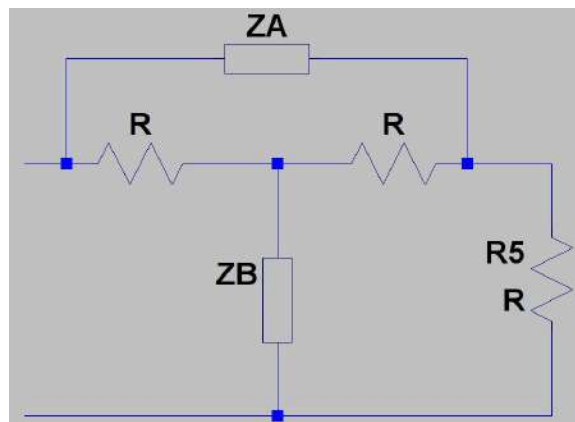
Apply MAI to obtain the Z_e of the following quadripole and verify by another method.



Exercise #5

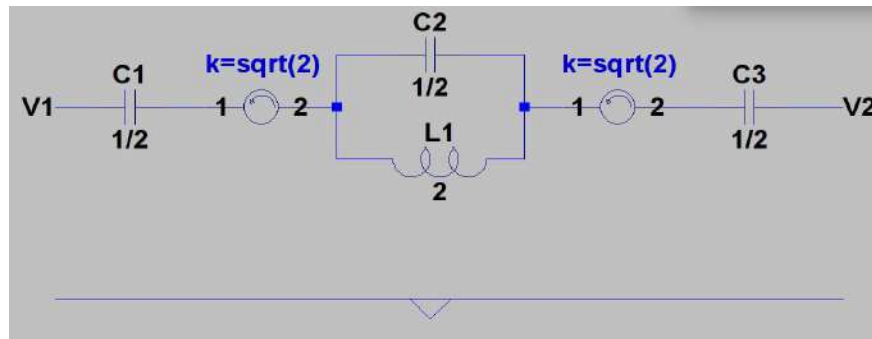
Given the following quadripole, apply MAI to obtain V_2/V_1 and Z_e

Note that $Z_a \times Z_b = R^2$



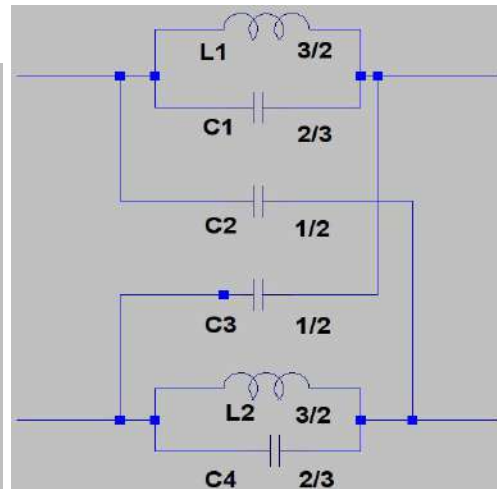
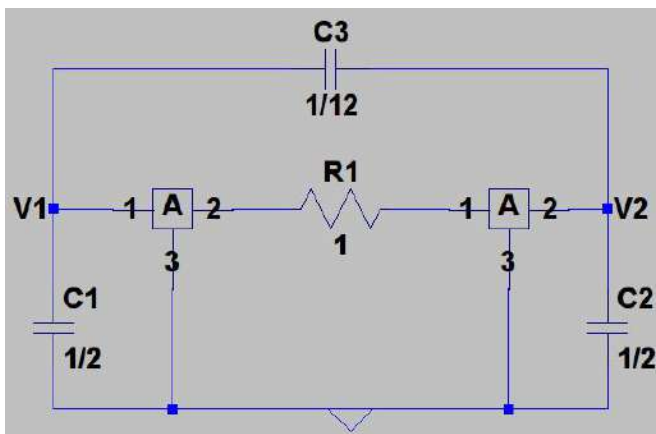
Exercise #6

Obtain an equivalent lattice circuit to the following quadripole



Exercise #7

Indicate how you would implement the “A” quadruples so that the active structures behave like the proposed lattices.



Exercise #8

The Y-parameters of a BS170 MOSFET operating under the following conditions are desired:

- $V_{DS} = 10V$
- $V_{GS} = 0V$
- $I_{DS} = 200mA$
- Common-source configuration.
- Operating frequency: 1 MHz

It is known that under these conditions the transistor presents the following characteristics (datasheet):

- $g_{fs} = 200 \text{ mS}$

- $C_{iss} = 35\text{pF}$
- $C_{rss} = 5\text{pF}$
- $C_{oss} = 32\text{pF}$

In addition, the manufacturer specifies the following:

- $C_{oss} = C_{ds} + C_{rss}$
- $C_{iss} = C_{gs} + C_{rss}$
- $C_{rss} = C_{gd}$

Find the Y-parameters of the transistor by setting $C_{rss} = 0$ and with the datasheet value.

Note: the datasheet can be downloaded here: <https://www.onsemi.com/pub/Collateral/BS170-D.PDF>

Exercise #9

Given the datasheets for the MPSH10 transistor and knowing the biasing conditions:

- $V_{CE} = 10\text{V}$
- $I_C = 2\text{mA}$

It is operated as a class A amplifier in a common-emitter configuration driven by the base with a 300 MHz sinusoidal signal. Required:

- Calculate the indefinite admittance matrix of the transistor and determine the Y-parameters matrix for the common-base configuration.
- Calculate the T-parameters matrix of the transistor in the common-emitter configuration.